

Class 13: RNASeq mini project

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Background

Today we will run through a complete RNASeq analysis.

The data comes from GEO entry: GSE37704, which is associated with the following publication:

Trapnell C, Hendrickson DG, Sauvageau M, Goff L et al. "Differential analysis of gene regulation at transcript resolution with RNA-seq". Nat Biotechnol 2013 Jan;31(1):46-53. PMID: 23222703

The authors report on differential analysis of lung fibroblasts in response to loss of the developmental transcription factor **HOXA1**. Their results and others indicate that HOXA1 is required for lung fibroblast and HeLa cell cycle progression. In particular their analysis show that "loss of HOXA1 results in significant expression level changes in thousands of individual transcripts, along with isoform switching events in key regulators of the cell cycle". For our session we have used their Sailfish gene-level estimated counts and hence are restricted to protein-coding genes only.

Data Import

```
counts <- read.csv("GSE37704_featurecounts.csv", row.names = 1)
metadata <- read.csv("GSE37704_metadata.csv")
```

Check correspondence of metadata and counts (i.e. that the columns in counts match the rows in the metadata)

```
metadata
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

```
head(counts)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

```
colnames(counts)
```

```
[1] "length"      "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370"
[7] "SRR493371"
```

```
metadata$id
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

Need to remove the length column in counts

```
test_cols <- !all(colnames(counts)[-1] == metadata$id)
test_cols
```

```
[1] FALSE
```

```
if (!test_cols) {
  print("Looks like you messed up")
}
```

```
[1] "Looks like you messed up"
```

```
counts <- counts[,-1]
```

Check it again:

```
all(colnames(counts) == metadata$id)
```

```
[1] TRUE
```

Also lets remove low count genes

```
tot.counts <- rowSums(counts)
head(tot.counts)
```

```
ENSG00000186092 ENSG00000279928 ENSG00000279457 ENSG00000278566 ENSG00000273547
                0                0                183                0                0
ENSG00000187634
                1129
```

So many genes with 0 counts so let's remove all zero count genes

```
zero.inds <- tot.counts == 0
head(zero.inds)
```

```
ENSG00000186092 ENSG00000279928 ENSG00000279457 ENSG00000278566 ENSG00000273547
                TRUE             TRUE             FALSE             TRUE             TRUE
ENSG00000187634
                FALSE
```

```
counts <- counts[!zero.inds,]
head(counts)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

Setup for DESeq

```
library(DESeq2)
```

```
dds <- DESeqDataSetFromMatrix(countData = counts,
                              colData = metadata,
                              design = ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

Run DESeq

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

Get Results

```
res <- results(dds)
```

```
head(res)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01
	padj				
	<numeric>				
ENSG00000279457	6.86555e-01				
ENSG00000187634	5.15718e-03				
ENSG00000188976	1.76549e-35				
ENSG00000187961	1.13413e-07				
ENSG00000187583	9.19031e-01				
ENSG00000187642	4.03379e-01				

Add Annotation

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res), # Our genenames
                     keytype="ENSEMBL",  # The format of our genenames
                     column="SYMBOL",     # The new format we want to add
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     column="ENTREZID",
                     keytype="ENSEMBL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$uniprot <- mapIds(org.Hs.eg.db,
                      keys=row.names(res),
                      column="UNIPROT",
                      keytype="ENSEMBL",
                      multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$genename <- mapIds(org.Hs.eg.db,
                       keys=row.names(res),
                       column="GENENAME",
                       keytype="ENSEMBL",
                       multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01

	padj	symbol	entrez	uniprot
	<numeric>	<character>	<character>	<character>
ENSG00000279457	6.86555e-01	NA	NA	NA
ENSG00000187634	5.15718e-03	SAMD11	148398	A0A087WX24
ENSG00000188976	1.76549e-35	NOC2L	26155	B3KNC3
ENSG00000187961	1.13413e-07	KLHL17	339451	Q5SV94
ENSG00000187583	9.19031e-01	PLEKHN1	84069	Q494U1
ENSG00000187642	4.03379e-01	PERM1	84808	Q5SV97

	genename
	<character>
ENSG00000279457	NA
ENSG00000187634	sterile alpha motif ..
ENSG00000188976	NOC2 like nucleolar ..
ENSG00000187961	kelch like family me..
ENSG00000187583	pleckstrin homology ..
ENSG00000187642	PPARGC1 and ESRR ind..

```
write.csv(res,file="myresults_annotated.csv")
```

You can arrange and view the results by the adjusted p-value

```
ord <- order( res$padj )
head(res[ord,])
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
--	----------	----------------	-------	------	--------

	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000117519	4483.63	-2.42272	0.0600016	-40.3776	0
ENSG00000183508	2053.88	3.20196	0.0724172	44.2154	0
ENSG00000159176	5692.46	-2.31374	0.0575534	-40.2016	0
ENSG00000116016	4423.95	-1.88802	0.0431680	-43.7366	0
ENSG00000164251	2348.77	3.34451	0.0690718	48.4208	0
ENSG00000124766	2576.65	2.39229	0.0617086	38.7675	0
	padj	symbol	entrez	uniprot	
	<numeric>	<character>	<character>	<character>	
ENSG00000117519	0	CNN3	1266	Q15417	
ENSG00000183508	0	TENT5C	54855	A3KMG2	
ENSG00000159176	0	CSRP1	1465	B3KVC9	
ENSG00000116016	0	EPAS1	2034	Q53SM6	
ENSG00000164251	0	F2RL1	2150	P55085	
ENSG00000124766	0	SOX4	6659	Q06945	
	genename				
	<character>				
ENSG00000117519	calponin 3				
ENSG00000183508	terminal nucleotidyl..				
ENSG00000159176	cysteine and glycine..				
ENSG00000116016	endothelial PAS doma..				
ENSG00000164251	F2R like trypsin rec..				
ENSG00000124766	SRY-box transcriptio..				

```
write.csv(res[ord,], "deseq_results.csv")
```

Visualize Results

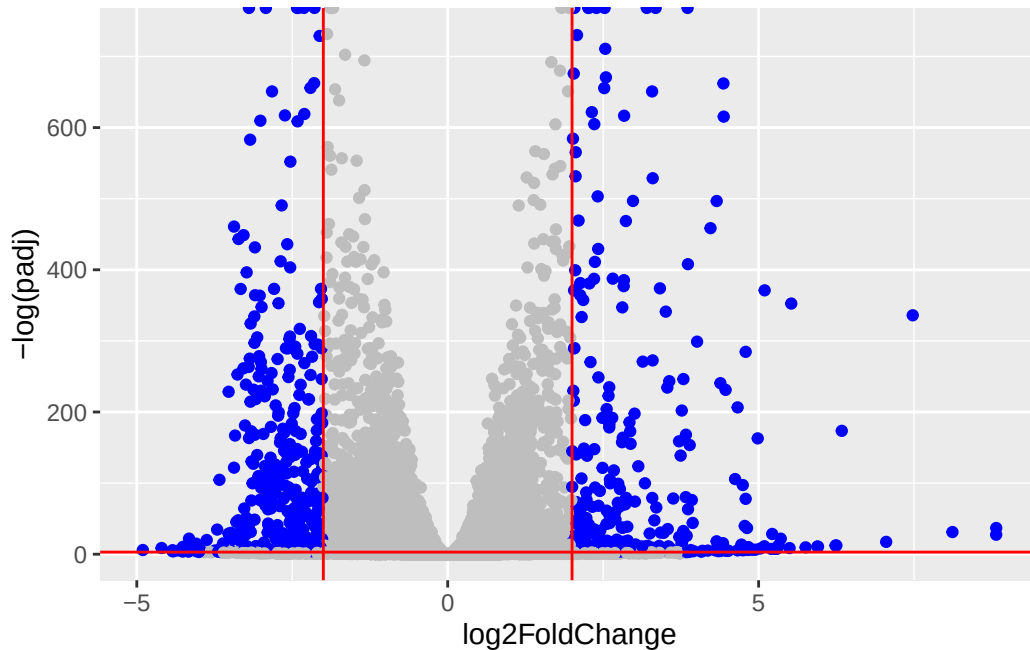
```
library(ggplot2)

mycols <- rep("gray", nrow(res))
mycols[abs(res$log2FoldChange)>2] <- "blue"
mycols[res$padj >= 0.05] <- "grey"

ggplot(res) + aes(log2FoldChange, -log(padj)) +
  geom_point(col=mycols)+
  geom_vline(xintercept = c(-2,2), col="red")+
  geom_hline(yintercept = -log(0.05), col = "red")
```

Warning: Removed 1237 rows containing missing values or values outside the scale range


```
(`geom_point()`).
```



Pathway Analysis

```
library(pathview)
```

```
#####  
Pathview is an open source software package distributed under GNU General  
Public License version 3 (GPLv3). Details of GPLv3 is available at  
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to  
formally cite the original Pathview paper (not just mention it) in publications  
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)
data(kegg.sets.hs)
```

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
<NA>      148398      26155      339451      84069      84808
0.17925708 0.42645712 -0.69272046 0.72975561 0.04057653 0.54281049
```

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
head(keggres$less, 3)
```

		p.geomean	stat.mean
hsa04110	Cell cycle	8.995727e-06	-4.378644
hsa03030	DNA replication	9.424076e-05	-3.951803
hsa05130	Pathogenic Escherichia coli infection	1.405864e-04	-3.765330
		p.val	q.val
hsa04110	Cell cycle	8.995727e-06	0.001889103
hsa03030	DNA replication	9.424076e-05	0.009841047
hsa05130	Pathogenic Escherichia coli infection	1.405864e-04	0.009841047
		set.size	exp1
hsa04110	Cell cycle	121	8.995727e-06
hsa03030	DNA replication	36	9.424076e-05
hsa05130	Pathogenic Escherichia coli infection	53	1.405864e-04

Looks like the top pathway is Cell Cycle ("hsa04110").

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/seona/Desktop/BGGN213/class13

Info: Writing image file hsa04110.pathview.png

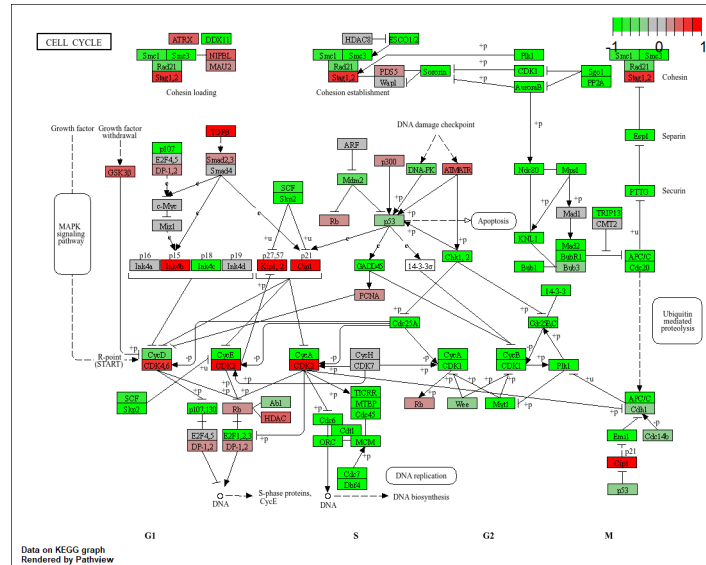


Figure 1: That pathway analysis

Let's try GO Analysis and compare to KEGG analysis

```
data(go.sets.hs)
data(go.subs.hs)
# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)
```

```
head(gobpres$less)
```

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059	chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236	mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10
		q.val	set.size	exp1
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15

G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059	chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236	mitotic prometaphase	1.178402e-07	84	1.729553e-10

```
lapply(gobpres, head)
```

\$greater

		p.geomean	stat.mean	p.val
G0:0007156	homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009	morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729	tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610	behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562	epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295	tube development	5.953254e-04	3.253665	5.953254e-04

		q.val	set.size	exp1
G0:0007156	homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009	morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729	tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610	behavior	0.1967577	426	1.925222e-04
G0:0060562	epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295	tube development	0.3565320	391	5.953254e-04

\$less

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059	chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236	mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10

		q.val	set.size	exp1
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059	chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236	mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

		stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.824205	3.824205
G0:0002009	morphogenesis of an epithelium	3.653886	3.653886

G0:0048729	tissue morphogenesis	3.643242	3.643242
G0:0007610	behavior	3.565432	3.565432
G0:0060562	epithelial tube morphogenesis	3.261376	3.261376
G0:0035295	tube development	3.253665	3.253665

Reactome Analysis

Some folks really like Reactome (i.e. their webpage viewer) rather than the R package of the same name (available from bioconductor).

To use the website viewer we want to upload our set of gene symbols for the genes we want to focus on (here those with a P-value ≤ 0.05).

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt",
            row.names=FALSE, col.names=FALSE, quote=FALSE)
```

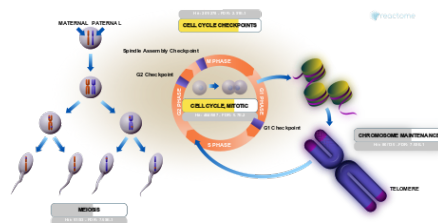


Figure 2: Result from Reactome

Save results

```
save(res, counts, metadata, file="my_results.RData")
```